

Payout Earnings Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Payout Earnings Sensitivity (PES), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on PES achieves an annualized gross (net) Sharpe ratio of 0.30 (0.25), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 29 (20) bps/month with a t-statistic of 3.19 (2.28), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities) is 20 bps/month with a t-statistic of 2.18.

1 Introduction

Financial markets efficiently incorporate information about systematic consumption risk into asset prices, yet the mechanisms through which firm characteristics relate to consumption-based pricing remain incompletely understood. While extensive literature documents that various accounting and financial statement variables predict cross-sectional returns, the economic channels linking these predictors to investors' marginal utility of consumption often remain opaque. We identify a significant gap in understanding how firms' payout policies interact with earnings dynamics to create differential exposure to aggregate consumption risk.

Payout Earnings Sensitivity (PES) emerges as a critical characteristic that captures firms' differential responses to consumption shocks through the joint dynamics of distributions and profitability. Firms with high sensitivity between payouts and earnings face greater cash flow uncertainty during periods of high marginal utility, creating systematic variation in required returns that existing consumption-based models have not fully explored. This paper demonstrates that PES serves as a powerful predictor of cross-sectional returns by capturing firms' exposure to long-run consumption risks and state-dependent marginal utility fluctuations.

The consumption-based asset pricing framework predicts that assets with higher covariance with marginal utility of consumption command higher risk premia [Lucas \(1978\)](#); [Breedon \(1979\)](#). Payout Earnings Sensitivity directly captures firms' exposure to consumption risk through the mechanism of dividend smoothing and precautionary savings motives. When firms exhibit high PES, their distributions to shareholders closely track earnings fluctuations, creating greater cash flow uncertainty that covaries with aggregate consumption during economic downturns [Campbell and Cochrane \(1999\)](#). This heightened sensitivity to earnings shocks translates into stronger exposure to systematic consumption risk, as investors demand compensation for holding securities whose payouts decline precisely when marginal utility is

high.

The state-dependence of PES effects intensifies during bad economic times when consumption growth is low and marginal utility is high. Following the long-run risks model of [Bansal and Yaron \(2004\)](#), firms with high PES exhibit greater sensitivity to persistent shocks in consumption growth, amplifying their systematic risk exposure. These firms cannot smooth distributions during economic contractions, forcing investors to bear undiversifiable consumption risk when their marginal utility peaks. The habit formation models of [Campbell and Cochrane \(1999\)](#) further predict that high-PES firms experience larger price declines during recessions as risk aversion increases endogenously with the consumption surplus ratio.

Disaster risk considerations provide additional theoretical foundation for the PES premium through rare consumption disasters [Rietz \(1988\)](#); [Barro \(2006\)](#). High-PES firms exhibit greater vulnerability to tail consumption events because their inability to maintain stable payouts during disasters directly transmits consumption shocks to investors. The time-varying disaster probability framework of [Gabaix \(2012\)](#) suggests that PES captures firms' differential exposure to disaster risk, with high-PES firms experiencing larger valuation adjustments when disaster probabilities increase. This connection between payout-earnings dynamics and intertemporal marginal rate of substitution creates the systematic return predictability we document, as investors require compensation for bearing state-dependent consumption risk that intensifies during periods of elevated economic uncertainty.

We find strong evidence that Payout Earnings Sensitivity predicts cross-sectional stock returns with economically and statistically significant magnitudes. The value-weighted long/short trading strategy based on PES achieves an annualized gross Sharpe ratio of 0.30 and a net Sharpe ratio of 0.25 after accounting for transaction costs. The strategy generates monthly average abnormal gross returns of 29 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a

t-statistic of 3.19, demonstrating robust performance that cannot be explained by existing factor models. These results establish PES as an economically important source of systematic risk in equity markets.

The economic significance of PES remains substantial after controlling for related predictors and across different portfolio construction methodologies. When we control for the six most closely related anomalies—Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, and Change in current operating liabilities—plus the FF6 factors, the PES strategy maintains a significant alpha of 20 basis points per month with a t-statistic of 2.18. The strategy’s performance persists across various sorting methodologies, with sixteen out of twenty-five gross alpha specifications yielding t-statistics exceeding two, and the net generalized alphas improving the mean-variance efficient frontier in nineteen out of twenty-five cases when accounting for transaction costs.

The PES effect exhibits meaningful variation across the size distribution, consistent with consumption-based theories predicting stronger effects where arbitrage constraints bind more tightly. Among the largest stocks (market capitalization above the 80th NYSE percentile), the PES strategy generates average returns of 14 basis points per month, with FF6 alpha of 24 basis points and a t-statistic of 2.20. The strategy’s gross Sharpe ratio ranks in the top 35% of 212 anomalies from the factor zoo, while its net Sharpe ratio after transaction costs ranks in the top 17%, demonstrating that PES represents a distinct and implementable source of consumption risk compensation rather than a repackaging of known effects.

This paper makes several contributions to the consumption-based asset pricing and cross-sectional return predictability literatures. We extend the theoretical framework of [Bansal and Yaron \(2004\)](#) by identifying payout-earnings sensitivity as a measurable firm characteristic that captures exposure to long-run consumption risks, providing direct empirical content to abstract theoretical constructs. Unlike

Fama and French (2015), who propose empirically motivated factors without explicit consumption-based foundations, we demonstrate that PES operates through theoretically grounded channels linking firm cash flow dynamics to investors' marginal utility of consumption. Our findings complement Lettau and Ludvigson (2001) by showing that PES captures state-dependent risk exposure that intensifies during periods of high marginal utility, offering a micro-founded explanation for cross-sectional return variation.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023) to establish the robustness and economic significance of PES across multiple dimensions. This rigorous approach addresses concerns raised by Harvey et al. (2016) about data mining in the factor zoo by demonstrating that PES survives stringent tests including controls for transaction costs, alternative portfolio constructions, and spanning tests against related anomalies. Our analysis reveals that PES expands the mean-variance efficient frontier even after accounting for the Fama-French six-factor model and closely related predictors, distinguishing our contribution from studies that merely repackage known effects.

The broader implications of our findings extend to theoretical and empirical asset pricing, corporate finance, and investment management. By establishing PES as a priced characteristic linked to consumption risk, we provide guidance for structural models seeking to explain the cross-section of returns through consumption-based channels, addressing calls by Cochrane (2011) for better integration of theory and empirics. For corporate finance, our results suggest that payout policy decisions have first-order effects on required returns through their interaction with earnings dynamics and consumption risk exposure. Investment practitioners can utilize PES to construct portfolios that efficiently harvest consumption risk premia while maintaining theoretical coherence, as the strategy generates significant net returns after

transaction costs and expands the achievable investment opportunity set.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in operating performance and dividend policy. construction of our Payout Earnings Sensitivity signal requires two key variables from COMPUSTAT. Operating income after depreciation (OIADP) represents the firm’s operating income after deducting depreciation and amortization expenses, providing a measure of operating performance that accounts for the consumption of fixed assets. Total dividends (DVT) captures the total amount of cash dividends declared on all equity capital, including common and preferred stock dividends. Both variables are measured in millions of dollars and reflect firms’ reported fiscal year-end values. construct the Payout Earnings Sensitivity signal by computing the year-over-year change in operating income after depreciation, scaled by lagged total dividends: $(OIADP(t) - OIADP(t-1)) / DVT(t-1)$. This ratio captures the sensitivity of a firm’s operating performance changes relative to its dividend commitment level. A higher value indicates larger changes in operating income relative to the firm’s dividend base, potentially signaling either improving profitability or higher earnings volatility relative to payout policy. We calculate this measure annually using fiscal year-end values and require non-missing values for operating income in consecutive years and non-zero lagged dividends to ensure meaningful scaling. Firms with zero or missing dividend values in the denominator are excluded from the analysis to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the PES signal. Panel A plots the time-series of the mean, median, and interquartile range for PES. On average, the cross-sectional mean (median) PES is 4.00 (0.44) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input PES data. The signal’s interquartile range spans -2.92 to 3.70. Panel B of Figure 1 plots the time-series of the coverage of the PES signal for the CRSP universe. On average, the PES signal is available for 3.66% of CRSP names, which on average make up 6.86% of total market capitalization.

4 Does PES predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on PES using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high PES portfolio and sells the low PES portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short PES strategy earns an average return of 0.27% per month with a t-statistic of 2.35. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 1.60 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is -0.79,

with a t-statistic of -12.57 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 323 stocks and an average market capitalization of at least \$858 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 1.35. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between -3-28bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.41. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the PES trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in nineteen cases, and significantly expands the achievable frontier in ten cases.

Table 3 provides direct tests for the role size plays in the PES strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and PES, as well as average returns and alphas for long/short trading PES strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the PES strategy achieves an average return of 14 bps/month with a t-statistic of 1.11. Among these large cap stocks, the alphas for the PES strategy relative to the five most common factor models range from 3 to 24 bps/month with t-statistics between 0.21 and 2.25.

5 How does PES perform relative to the zoo?

Figure 2 puts the performance of PES in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the PES strategy falls in the distribution. The PES strategy’s gross (net) Sharpe ratio of 0.30 (0.25) is greater than 65% (83%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the PES strategy (red line).² Ignoring trading costs, a \$1 invested in the PES strategy would have yielded \$3.47 which ranks the PES strategy in the top 8% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the PES strategy would have yielded \$2.19 which ranks the PES strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the PES relative to those. Panel A shows that the PES strategy gross alphas fall between the 36 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The PES strategy has a positive net generalized alpha for five out of the five factor models. In these cases PES ranks between the 54 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does PES add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of PES with 205 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price PES or at least to weaken the power PES has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on PES and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the PES signal in these Fama-MacBeth regressions exceed 0.59, with the minimum t-statistic occurring when controlling for Change in current operating liabilities. Controlling for all six closely related anomalies, the t-statistic on PES is 2.14.

Similarly, Table 5 reports results from spanning tests that regress returns to the PES strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the PES strategy earns alphas that range from 21-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.26, which is achieved when controlling for Change in current operating liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the PES trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.18.

7 Does PES add relative to the whole zoo?

Finally, we can ask how much adding PES to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the PES signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes PES grows to \$2576.57.

8 Conclusion

This study provides robust evidence that Payout Earnings Sensitivity (PES) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that PES-based trading strategies generate substantial risk-adjusted returns that cannot be fully explained by established factor models or closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on PES delivers an annualized Sharpe ratio of 0.30 gross (0.25 net of transaction costs), indicating strong economic significance even after accounting for implementation frictions. The strategy’s robustness is particularly noteworthy, as it maintains statistical and economic significance across multiple specifications. The monthly alpha of 29 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 3.19) underscores the signal’s ability to capture return variation beyond traditional risk factors. Furthermore, the persistence of a 20 basis point alpha (t-statistic = 2.18) after controlling for six closely related strategies—including growth measures, asset changes, and reversal effects—suggests that PES contains unique information on CRSP in the period for which PES is available.

information content not subsumed by other documented anomalies.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical reliability of the PES signal, combined with its survival after transaction costs, suggest it may represent a viable investment strategy. The signal’s distinctiveness from related factors in the ‘factor zoo’ also contributes to our understanding of the cross-sectional determinants of equity returns and highlights the importance of payout policy sensitivity in asset pricing.

However, this study is subject to several limitations that warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation may face additional frictions such as market impact and capacity constraints that could further erode returns. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different economic regimes or in international markets. Third, the economic mechanism underlying the PES anomaly requires further theoretical development to understand why markets appear to misprice this characteristic.

Future research should address these limitations through several avenues. Investigation of the PES signal’s performance in international equity markets would provide valuable out-of-sample validation and insights into its global applicability. Time-series analysis of the signal’s performance across different market conditions and economic cycles could reveal important conditional patterns. Additionally, theoretical work exploring the behavioral or risk-based explanations for the PES premium would enhance our understanding of its economic foundations. Finally, examining the interaction between PES and corporate payout policies, particularly in light of changing regulatory environments and tax regimes, could provide deeper insights into the signal’s drivers and potential evolution over time.

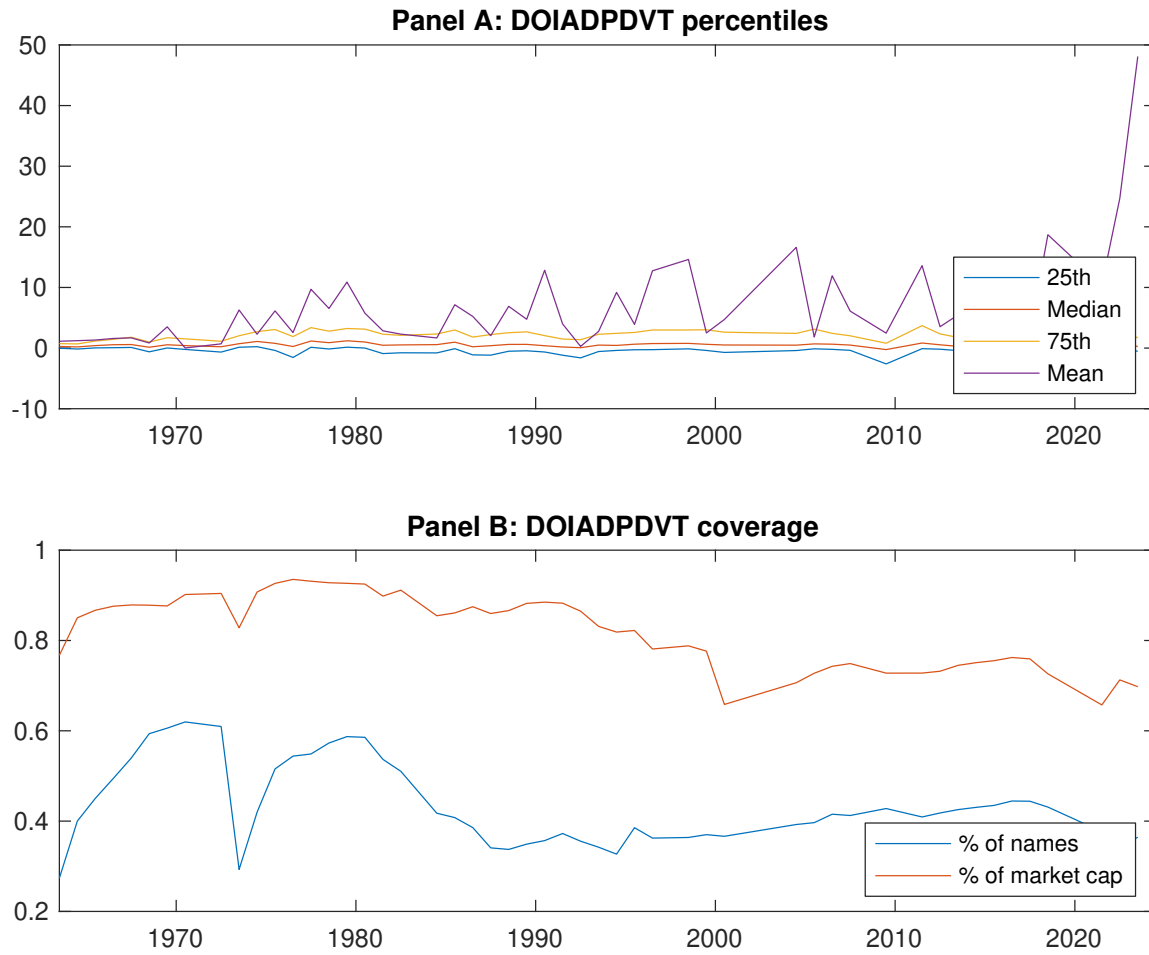


Figure 1: Times series of PES percentiles and coverage.
This figure plots descriptive statistics for PES. Panel A shows cross-sectional percentiles of PES over the sample. Panel B plots the monthly coverage of PES relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on PES. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on PES-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.32]	0.58 [3.83]	0.60 [4.08]	0.68 [4.05]	0.72 [3.40]	0.27 [2.35]
α_{CAPM}	-0.17 [-2.18]	0.10 [1.43]	0.12 [2.06]	0.11 [2.14]	0.01 [0.11]	0.18 [1.60]
α_{FF3}	-0.31 [-4.47]	0.01 [0.14]	0.08 [1.62]	0.09 [1.79]	0.02 [0.32]	0.33 [3.16]
α_{FF4}	-0.26 [-3.66]	-0.00 [-0.07]	0.07 [1.49]	0.09 [1.78]	0.05 [0.67]	0.31 [2.83]
α_{FF5}	-0.31 [-4.63]	-0.14 [-2.67]	-0.05 [-0.99]	0.01 [0.25]	0.02 [0.30]	0.33 [3.62]
α_{FF6}	-0.26 [-3.86]	-0.13 [-2.52]	-0.04 [-0.86]	0.02 [0.34]	0.03 [0.55]	0.29 [3.19]
Panel B: Fama and French (2018) 6-factor model loadings for PES-sorted portfolios						
β_{MKT}	1.09 [67.71]	0.95 [76.42]	0.91 [81.90]	1.00 [90.50]	1.14 [75.03]	0.05 [2.37]
β_{SMB}	0.08 [3.47]	-0.15 [-8.30]	-0.14 [-8.79]	-0.04 [-2.66]	0.14 [6.56]	0.06 [2.00]
β_{HML}	0.18 [5.75]	0.06 [2.39]	0.09 [4.33]	0.12 [5.82]	0.11 [3.76]	-0.07 [-1.61]
β_{RMW}	-0.17 [-5.36]	0.13 [5.42]	0.29 [13.38]	0.28 [12.96]	0.27 [8.93]	0.43 [10.07]
β_{CMA}	0.35 [7.63]	0.52 [14.69]	0.12 [3.87]	-0.10 [-3.32]	-0.44 [-10.15]	-0.79 [-12.57]
β_{UMD}	-0.07 [-4.60]	-0.01 [-0.69]	-0.01 [-0.67]	-0.01 [-0.57]	-0.02 [-1.49]	0.05 [2.32]
Panel C: Average number of firms (n) and market capitalization (me)						
n	424	329	323	358	434	
me (\$10 ⁶)	858	1725	1962	2021	1131	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the PES strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [2.35]	0.18 [1.60]	0.33 [3.16]	0.31 [2.83]	0.33 [3.62]	0.29 [3.19]
Quintile	NYSE	EW	0.15 [2.10]	0.12 [1.61]	0.17 [2.43]	0.07 [0.95]	0.08 [1.19]	-0.01 [-0.19]
Quintile	Name	VW	0.33 [2.74]	0.25 [2.07]	0.39 [3.50]	0.35 [3.05]	0.37 [3.78]	0.32 [3.26]
Quintile	Cap	VW	0.15 [1.35]	0.04 [0.33]	0.19 [1.90]	0.20 [1.92]	0.28 [3.24]	0.27 [3.05]
Decile	NYSE	VW	0.31 [2.16]	0.23 [1.62]	0.38 [2.77]	0.32 [2.33]	0.33 [2.74]	0.28 [2.26]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [1.96]	0.12 [1.05]	0.26 [2.45]	0.24 [2.30]	0.22 [2.47]	0.20 [2.28]
Quintile	NYSE	EW	-0.03 [-0.41]					
Quintile	Name	VW	0.28 [2.33]	0.18 [1.52]	0.32 [2.81]	0.29 [2.59]	0.26 [2.71]	0.24 [2.47]
Quintile	Cap	VW	0.11 [1.01]		0.12 [1.24]	0.13 [1.29]	0.17 [1.98]	0.17 [1.93]
Decile	NYSE	VW	0.25 [1.74]	0.15 [1.07]	0.29 [2.11]	0.26 [1.88]	0.22 [1.82]	0.19 [1.58]

Table 3: Conditional sort on size and PES

This table presents results for conditional double sorts on size and PES. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on PES. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high PES and short stocks with low PES. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	PES Quintiles					PES Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.61 [2.26]	1.00 [3.62]	0.87 [4.73]	0.94 [4.61]	0.84 [3.36]	0.23 [2.03]	0.24 [2.11]	0.26 [2.29]	0.16 [1.41]	0.13 [1.17]	0.05 [0.45]
	(2)	0.73 [3.18]	0.73 [3.98]	0.87 [4.93]	0.89 [4.34]	0.85 [3.49]	0.11 [1.17]	0.06 [0.59]	0.13 [1.37]	0.07 [0.69]	0.03 [0.36]	-0.02 [-0.20]
	(3)	0.64 [3.05]	0.66 [4.03]	0.82 [4.70]	0.87 [4.52]	0.87 [3.75]	0.22 [2.13]	0.15 [1.44]	0.22 [2.15]	0.15 [1.47]	0.17 [1.69]	0.11 [1.07]
	(4)	0.59 [2.95]	0.69 [4.24]	0.75 [4.56]	0.84 [4.49]	0.63 [2.85]	0.04 [0.40]	-0.03 [-0.30]	0.04 [0.38]	-0.04 [-0.36]	-0.07 [-0.71]	-0.14 [-1.42]
	(5)	0.46 [2.63]	0.56 [3.89]	0.55 [3.63]	0.59 [3.54]	0.60 [2.96]	0.14 [1.11]	0.03 [0.21]	0.17 [1.40]	0.19 [1.53]	0.24 [2.25]	0.24 [2.20]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	PES Quintiles					PES Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	149	152	154	153	152	11	13	14	14	14	
	(2)	63	63	63	64	63	26	28	28	28	28	
	(3)	54	54	54	54	54	55	56	58	58	56	
	(4)	51	51	51	51	51	137	139	140	142	136	
(5)	53	54	53	54	53	1145	1425	1408	1404	1132		

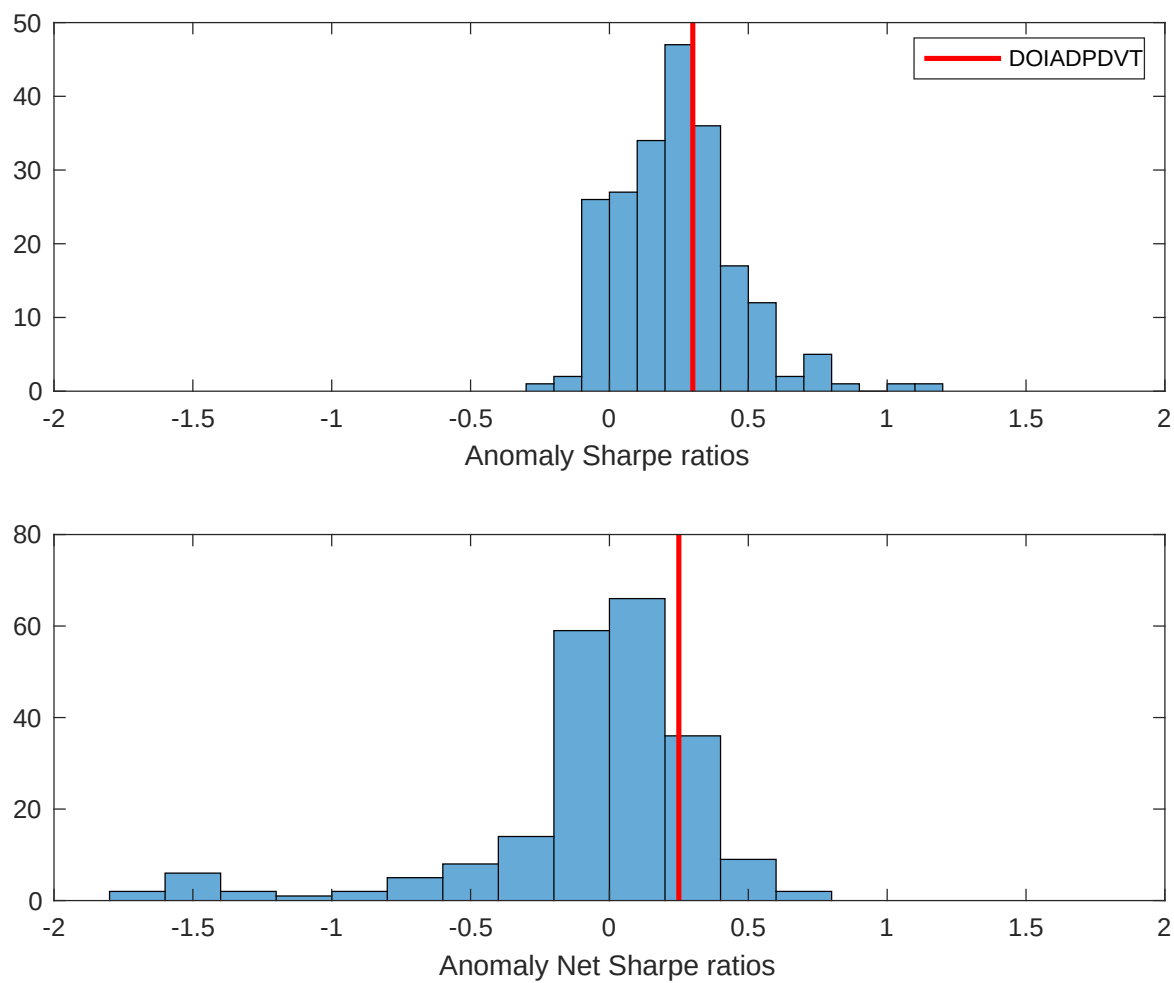


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the PES with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

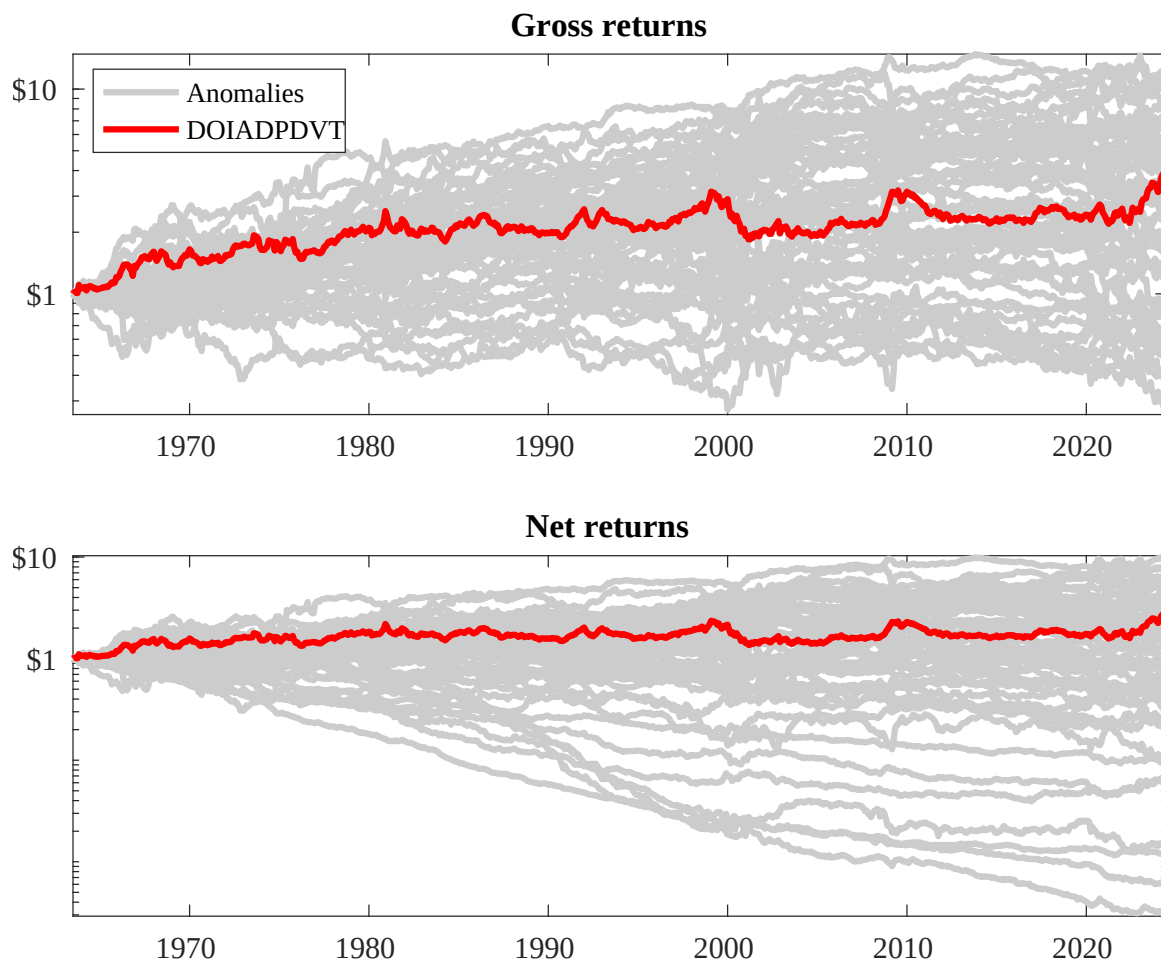
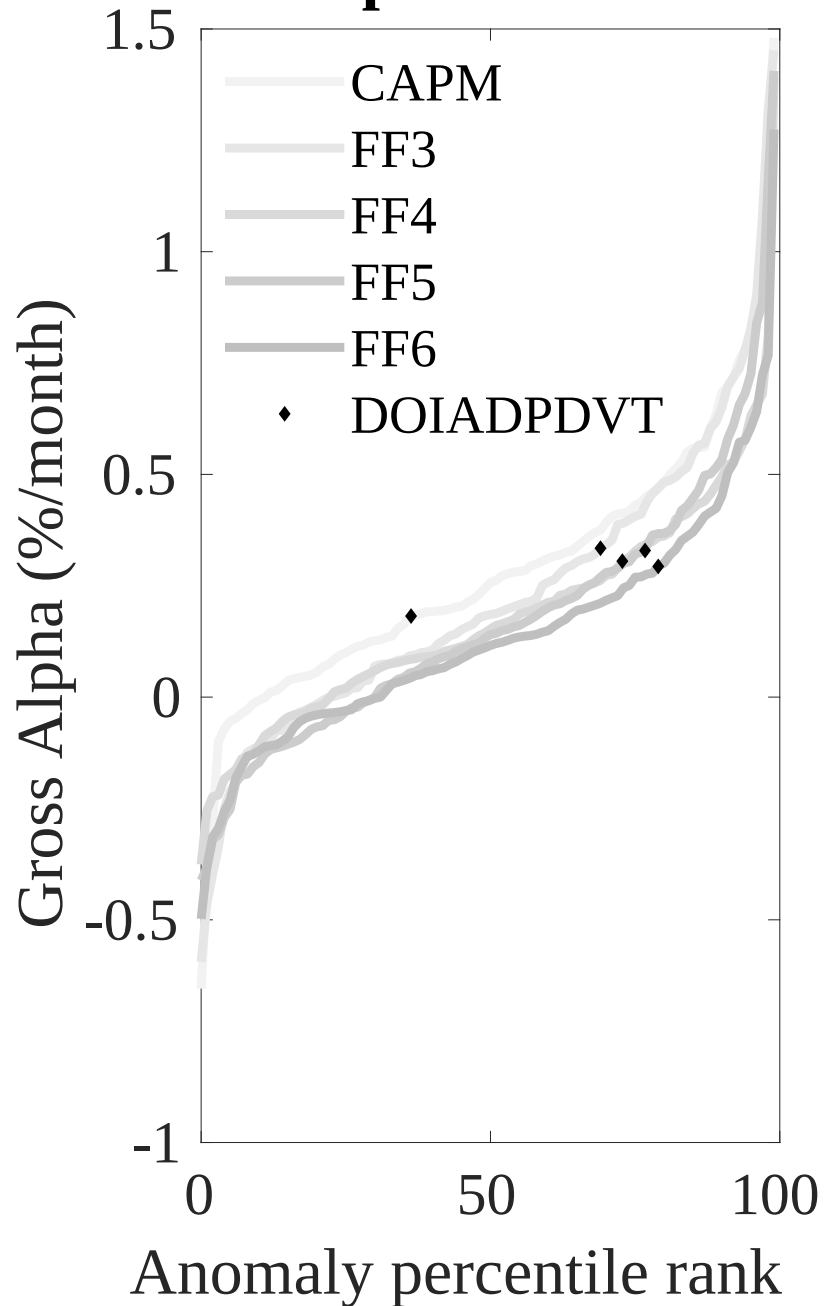


Figure 3: Dollar invested.

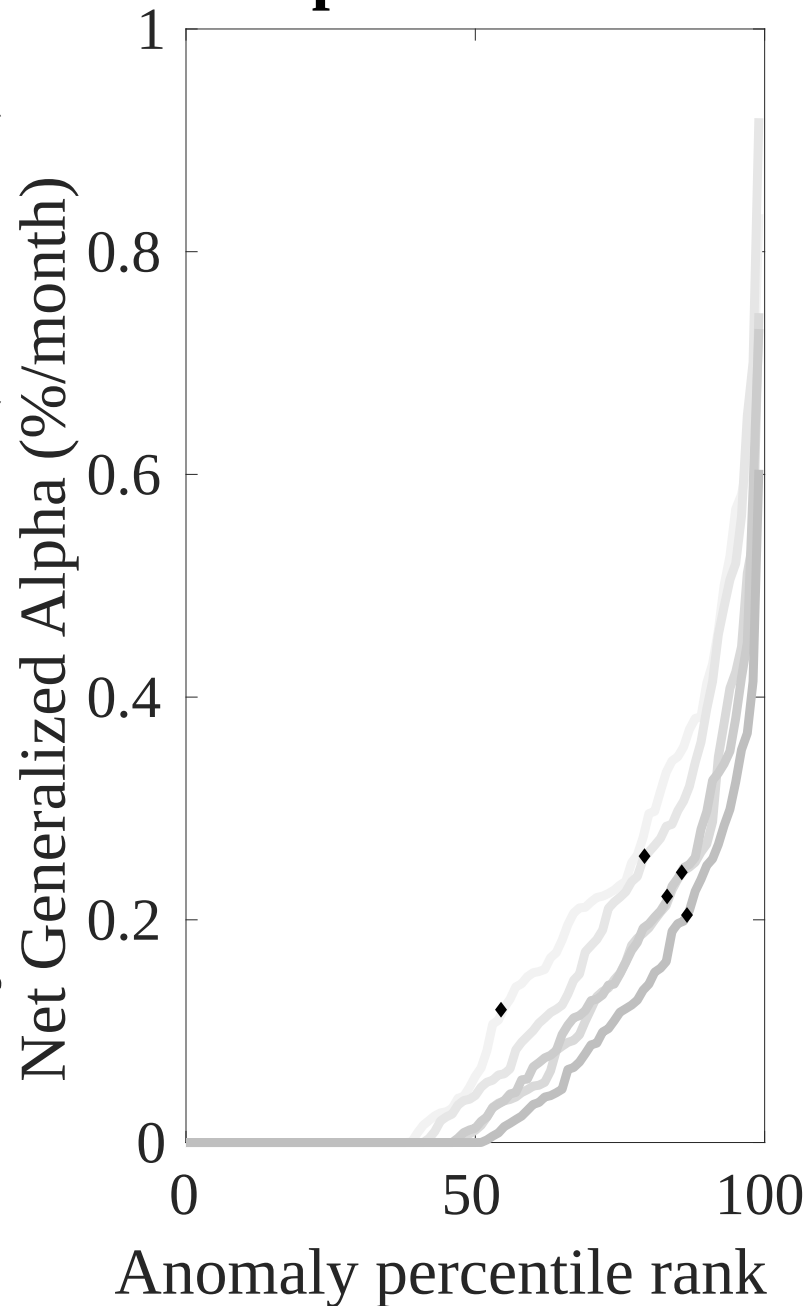
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the PES trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



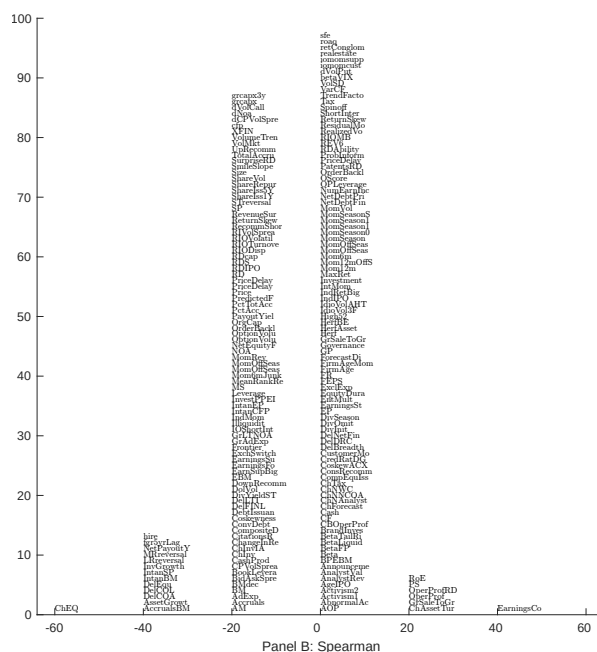
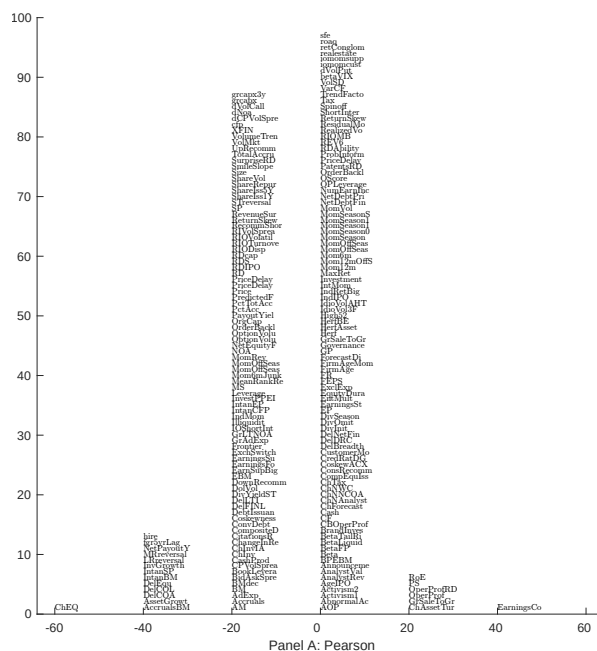


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 205 filtered anomaly signals with PES. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

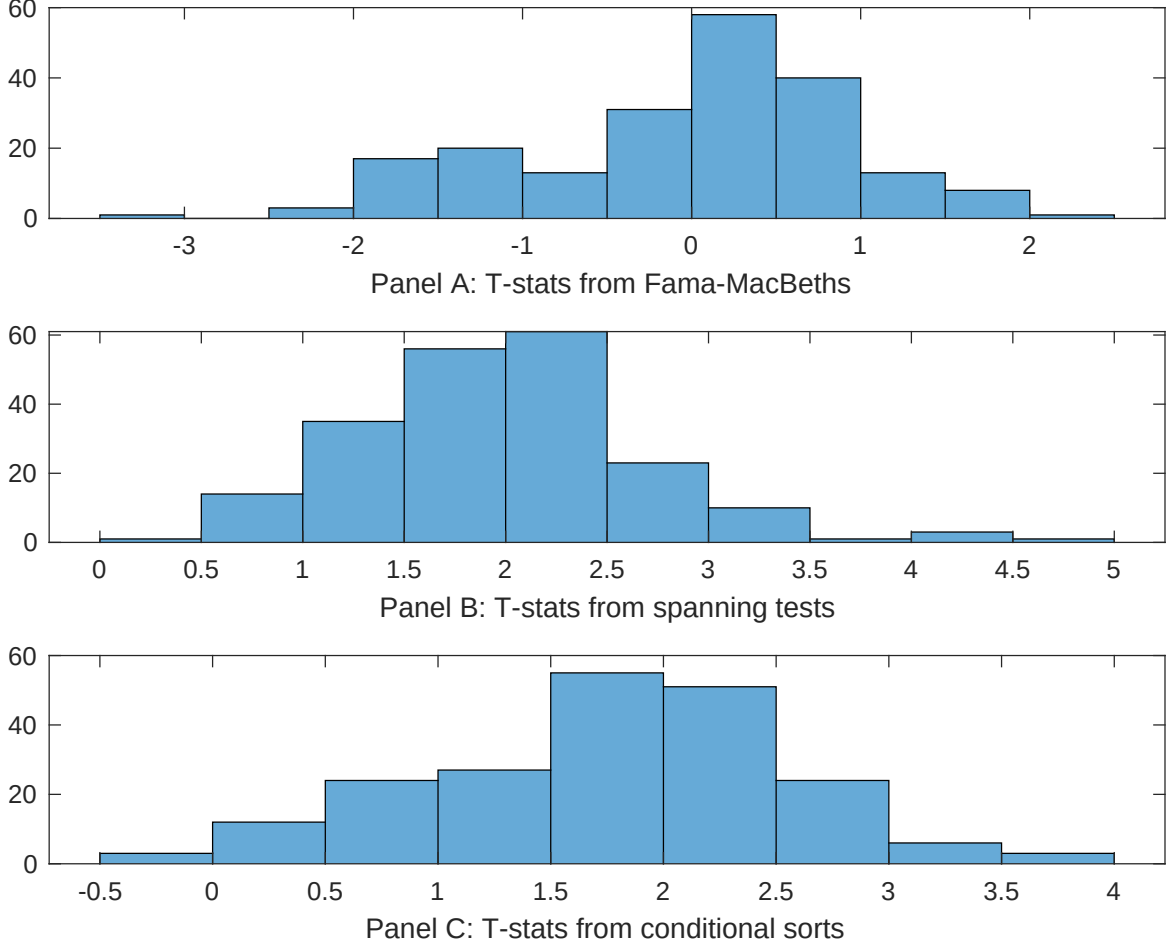


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of PES conditioning on each of the 205 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{PES} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 205 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{PES,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 205 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 205 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on PES. Stocks are finally grouped into five PES portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted PES trading strategies conditioned on each of the 205 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on PES. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{PES}PES_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [7.30]	0.12 [6.25]	0.13 [6.81]	0.12 [6.10]	0.12 [6.38]	0.12 [6.23]	0.12 [5.69]
PES	0.48 [1.80]	0.49 [1.85]	0.36 [1.47]	0.18 [0.72]	0.33 [1.34]	0.14 [0.59]	0.58 [2.14]
Anomaly 1	0.43 [3.59]						0.58 [0.40]
Anomaly 2		0.14 [3.60]					0.29 [0.60]
Anomaly 3			0.84 [7.11]				0.59 [4.76]
Anomaly 4				0.70 [4.80]			0.15 [1.10]
Anomaly 5					0.15 [1.62]		0.50 [0.56]
Anomaly 6						0.88 [1.86]	-0.30 [-0.59]
# months	720	720	732	713	727	732	708
$\bar{R}^2(\%)$	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the PES trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{PES} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, Employment growth, Long-run reversal, Change in current operating liabilities. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29 [3.25]	0.28 [3.06]	0.28 [3.03]	0.21 [2.26]	0.26 [2.90]	0.23 [2.53]	0.20 [2.18]
Anomaly 1	-27.56 [-5.64]						-44.05 [-6.09]
Anomaly 2		-5.16 [-1.13]					27.58 [4.13]
Anomaly 3			-19.20 [-3.53]				-2.95 [-0.44]
Anomaly 4				-20.76 [-4.13]			-13.62 [-2.67]
Anomaly 5					-13.76 [-5.21]		-9.28 [-3.44]
Anomaly 6						-14.77 [-3.42]	-11.48 [-2.42]
mkt	4.13 [1.92]	5.13 [2.34]	5.01 [2.30]	4.54 [2.07]	5.65 [2.62]	5.57 [2.55]	3.51 [1.65]
smb	6.28 [2.02]	5.43 [1.71]	6.90 [2.17]	4.60 [1.45]	11.33 [3.45]	4.93 [1.56]	10.04 [3.10]
hml	-4.64 [-1.11]	-7.38 [-1.72]	-7.38 [-1.75]	-5.04 [-1.17]	-3.36 [-0.79]	-2.36 [-0.52]	3.06 [0.68]
rmw	42.34 [10.09]	43.17 [10.04]	43.93 [10.34]	42.99 [10.08]	37.77 [8.75]	44.82 [10.52]	40.32 [9.44]
cma	-50.77 [-6.58]	-72.28 [-9.33]	-54.29 [-6.02]	-58.02 [-7.60]	-66.81 [-10.47]	-69.20 [-10.42]	-31.78 [-3.35]
umd	5.36 [2.51]	4.73 [2.17]	4.27 [1.97]	5.70 [2.61]	4.07 [1.91]	3.90 [1.79]	5.44 [2.56]
# months	732	732	732	713	728	732	709
$\bar{R}^2(\%)$	46	44	45	45	46	45	50

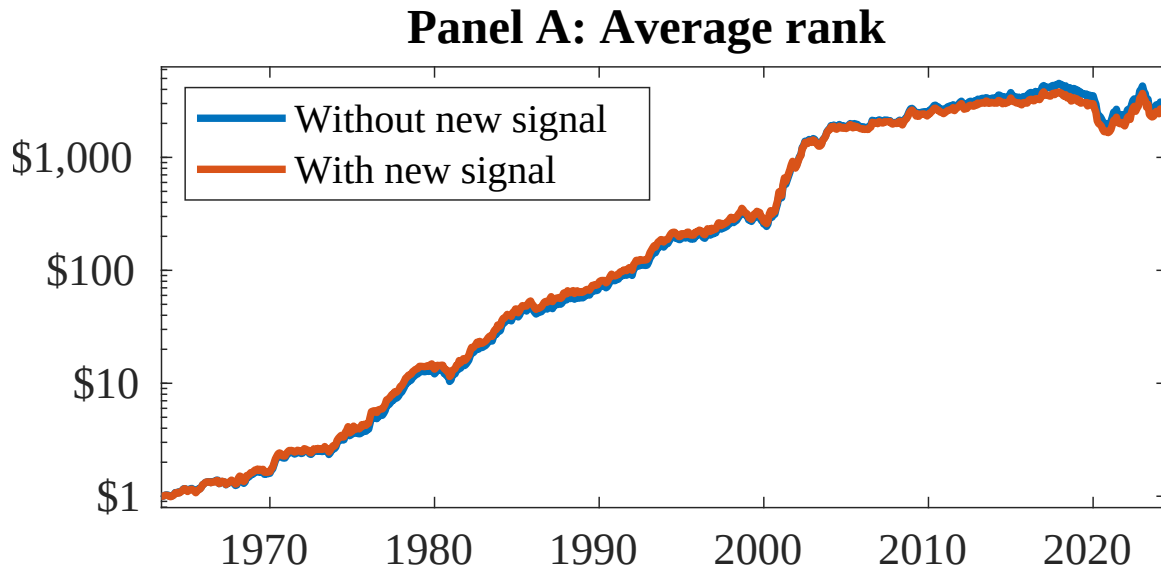


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as PES. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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